

Measuring Definitions: When Is a Definition Good, and ?How Would We Know

version 6 · corpus: 372 cases (289 positive · 83 negative) across 34 papers · 2026-08-09

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Abstract

A researcher who proposes a definition of a concept — "art", "consciousness", "game" — cannot say how good it is; they can only argue for it. The philosophical literature has offered criteria for the quality of a definition for seventy years, but no one has turned those criteria

into a number measured against a body of evidence. We present a method that does. A definition is treated as a binary classifier over cases **the literature itself has adjudicated**; every adjudication carries a verified verbatim quotation; and rival definitions are ranked in a single run. Before any scoring, definitions pass three logical gates that no statistical measure detects — circularity, recursion without a base case, and a free parameter its author never fixed. We report inter-coder reliability on the act of application itself ($\kappa = 0.547$, $n = 140$), a measurement almost absent from the interpretive-tagging work we surveyed. The method returns one unambiguous ranking from a single command, and states explicitly which definitions cannot be ranked and why. The primary aim is practical: a user brings a corpus and .a set of rival definitions, and receives the best definition that corpus supports

The need .1

Suppose you are a researcher and you must define a concept. Five rival definitions exist in the ?literature, and you have a sixth. How do you determine which is better

The available instruments were built for a different task. Philosophical analysis establishes *which* conditions are defensible; corpus linguistics measures *usage*. Neither returns a score for a given definition against a body of evidence. Comparison between candidates therefore proceeds by argument and counterexample — an excellent way to sharpen a position, and a poor way to choose .between positions. **What is missing is a number**

The need is not confined to philosophy. It arises in every project that builds a tagged corpus: the moment one fixes which items fall under a category, a definition has been applied — usually without being stated, and always without being measured. A tool that returns a score makes that .decision visible and checkable

The operational requirement, in one line: a single score between 0 and 1 for how good a given definition is, given a given corpus. Not a set of considerations — **a number**, reported .alongside what it does not cover

What the literature already agrees on .2

Agreement is broader than it appears, and it divides in two: **what can be measured**, and **what is a position in a debate**

What can be measured

The formal account of a well-formed definition was set out by **Suppes (1957)** and sharpened by **Belnap (1993)** into two conditions: **eliminability** — every sentence containing the defined term can be restated without it; and **conservativeness** — the definition licenses nothing new that does not concern the term itself. The Stanford Encyclopedia (**Gupta & Mackereth, 2023**) names what we in fact measure: **extensional adequacy** — whether the definition admits exactly what it should admit

Belnap's two conditions are **logical gates**: they are settled yes/no before any scoring. Extensional adequacy is what is measured continuously, and it divides immediately into two complementary quantities: **coverage** — how much of what should be admitted is admitted; and **discrimination** — .how much of what should be excluded is excluded

Here is the translation the whole method rests on. **Low coverage refutes necessity**: the definition contains a condition that is not necessary, and it excludes cases that belong. **Low discrimination refutes sufficiency**: the conditions are not sufficient, and cases that are not .the concept are admitted. Two binary logical notions become two continuous measures

What is a position in a debate

Against this stands the anti-essentialist tradition. **Wittgenstein (1953)** argued that "game" is organised not by conditions but by a network of family resemblance with blurred edges. **Weitz (1956)** extended this to art and argued that it is an **open concept** that must not be closed; **Kennick (1958)** added that traditional aesthetics erred in seeking a definition at all, not in the definitions it .proposed. **Ziff (1953)** had already formulated the task

The modern form of that position is the **cluster concept**: **Gaut (2000, 2005)** offers ten criteria of which none is necessary, and **Dutton (2006)** builds a naturalistic list of recognition criteria. In parallel, procedural definitions developed: **Danto (1964)** and **Dickie (1974, 1984, 2000)** locate art status in an institution rather than a property; **Levinson (1979)** locates it in a relation to prior art; .and **Carroll (1993)** replaces definition with historical narrative

The distinction that organises the whole field, and is easily missed: "necessary and sufficient conditions" is **a form of definition, not a criterion of its quality**. The dispute between Wittgenstein and those who define by conditions is not about what makes a definition good; it is about what **form** a definition should take. The criterion set must therefore be **form-neutral**. Otherwise a cluster concept and a condition-based concept cannot appear in the same table, and the ranking would measure form rather than quality.

What is still missing .3

None of these positions is formulated so that it can be scored against evidence. They state *what* a definition should do, not *how much* of it a given definition does.

Adjacent fields are in the same position. In natural language processing, the *definition modelling* task scores a generated definition by its similarity to a dictionary gloss, so a novel and correct definition is penalised exactly as a wrong one is. Ontology evaluation measures structure and consistency, not fit to usage. Standard scoring methods for classification — MCC (**Matthews, 1975; Chicco & Jurman, 2020**), inter-coder reliability (**Cohen, 1960; Krippendorff, 2018**), and bootstrap confidence intervals (**Efron, 1979**) — exist and are available, but have not been turned on this question.

.The gap is not an absence of criteria. It is an absence of measurement

This has a concrete cost. Weitz's claim that art is an open concept is, in its form, an **empirical** claim: if it is true, every condition-based definition must fail on coverage or on discrimination against a real corpus. In seventy years no one has checked it in numbers.

What the method does .4

A definition is treated as a **binary classifier** over cases the corpus itself has adjudicated: a case is positive if a paper states explicitly that it is the concept, and negative if a paper states explicitly that it is not. Silence is not negation, and every adjudication carries a verified verbatim quotation.

Before scoring, definitions pass **three gates** — circularity, recursion without a base case, and a free parameter the author never fixed. A definition that fails one is reported and left unranked

Survivors are ranked by **MCC** rather than F_1 , because a positively skewed corpus rewards a classifier that says yes to everything; ties are settled by a **paired bootstrap at the level of the paper**, because cases from one paper constitute a single voice. Alongside the ranking we report the **reliability of application**: κ between two independent coders

The output is one list: place, score, confidence interval — and beside it the definitions that were disqualified, with the reason

What the method is not .5

- **It does not adjudicate what art is**, or any other concept. The adjudications are the literature's; the method counts them
- **It does not validate the philosophical literature and does not replace it.** The primary aim is practical: a user with a corpus and rival definitions receives the best among them. Validating definitions in the literature is a possible use, and not the one it was built for
- **A definition that wins here has won on one corpus.** The corpus is a parameter of the measurement, not its background: a different corpus returns a different number, and that is correct rather than a defect
- **It does not score the Wittgensteinian position itself.** "Do not define by conditions" is not a rival definition but a proposal to forgo definition, and so falls outside the instrument's range — a stated limitation, not a concealed one
- **It does not read for the researcher.** The method narrows the field of candidates and shows where each one breaks; what to do about that remains a human decision

The three gates .6

A definition is a promise: it says what is admitted and what is excluded. It can be tested only where cases already have an adjudication — and here the adjudication is not ours but **the literature's**. A case is positive if a paper states explicitly that it is the concept, negative if a paper states explicitly that it is not. Silence is not negation. Every case carries a verified verbatim quotation

What the gates disqualify, and why before scoring rather than after

Three definitions in the corpus scored highly, and in each a defect emerged that no statistical measure detects. They are therefore disqualified **before** scoring and left unranked

Gate 1 — circularity

The definiens contains the definiendum. Our circular control ("whatever people call art") achieves an almost perfect MCC precisely because it copies the answer. No measure will ever catch it

Gate 2 — recursion without a base case

Levinson's historical definition: "a thing intended for regard as prior artworks were correctly regarded." It defines art by reference to prior art and never states what the first art was. The mathematical defence (0 is a number; $n+1$ is a number) works only when the base is given. Here there is $n+1$ and no 0

Gate 3 — failure of point identification

Gaut's cluster account: an object is art if it meets **enough** of ten criteria, none of which is necessary — and he never says how many "enough" is. Dutton, who defends the same approach, concedes the point explicitly: "*A recognition-criteria approach ... does not tell us in advance how many of the criteria need be present*"

We decomposed the ten criteria and coded 120 cases against each. Moving the threshold from $k = 1$ to $k = 10$ moves the score across a range of **0.6** MCC, and if k may be chosen per case the account attains MCC 1.000. **A definition with a free parameter is not falsifiable, and its score measures whoever applies it rather than the corpus**

Why the gates precede scoring

All three share one property: **no statistical measure sees them**. A circular definition classifies excellently precisely because it copies the answer; a definition with a free threshold receives whatever score its applier selected for it. If the gate runs after scoring, the defective definition already sits at the top of the table and has to be struck out afterwards — which is exactly what happened to us once, when an automated search was allowed to build a definition from a criterion

Measurement versus refinement — a distinction that must be recorded

A definition changes when a case defeats it, so every definition carries a version history. But one
:difference must not be blurred

.Measurement — the wording was fixed *before* the cases were seen. The score is a test

Refinement — the wording was fixed *after* it became clear which cases failed. The score is not a
.test, because the cases are already known

Both are legitimate, and neither may be reported as though the two were measured under the same
.conditions. The apparatus retains every version, never deletes, and marks each one accordingly

Sub-concepts must themselves be defined

A definition resting on an undefined term transfers the adjudication to unwritten judgement. In a
case we measured, defining the internal term according to what the corpus actually does with it —
without altering a single word of the definition itself — raised the score from +0.380 to +0.664. **A**
definer does not repair their own terms; those are settled outside the definition. The definition
.therefore unfolds recursively until further decomposition stops changing the score

Criteria for a good definition .7

Three are measured inside the confusion matrix, three outside it, and three gates precede all of
.them

Where measured	What it asks	Criterion	#
gate	?Does the definiens contain the definiendum	Non-circular	G1
gate	?If recursive — is the base case stated	Recursive base	G2
gate	?Does it contain a parameter the author never fixed	Point-identified	G3

Where measured	What it asks	Criterion	#
recall	How much of what is agreed to be the concept does it ?admit	Coverage	1
precision	How much of what is agreed not to be the concept ?does it exclude	Discrimination	2
held-out sample	?Does it hold on cases not used in building it	Prediction	3
rank correlation	Does it explain why one instance is more central than ?another	Typicality ranking	4
κ between coders	?Will two people reach the same answer	Applicability	5
ablation	?Can a condition be removed with nothing changing	Minimality	6

Criterion 5 is a precondition, not one more row. Without agreement between two coders on what counts as a true positive, the four cells of the matrix are meaningless. On a controlled layer of statistical terms it has been measured ($\kappa = 0.767$, "substantial"); for definitions it is .measured in §10

The confusion matrix .8

Here the "classifier" is the definition, and "ground truth" is **what the papers themselves state**. This is not absolute truth; it is the truth the corpus supplies, and that is precisely what makes the .measurement relative to a corpus

The four cells

- .TP** — the definition admits a case the papers declare to be the concept •

$$TP = |\{ x : D(x)=1 \text{ AND } \text{corpus}(x)=\text{yes} \}|$$

- .FP** — the definition admits a case the papers declare not to be the concept •

$$FP = |\{ x : D(x)=1 \text{ AND } \text{corpus}(x)=\text{no} \}|$$

• **.FN** — the definition excludes a case the papers declare to be the concept

$$FN = |\{ x : D(x)=0 \text{ AND } \text{corpus}(x)=\text{yes} \}|$$

• **.TN** — the definition excludes a case the papers declare not to be the concept

$$TN = |\{ x : D(x)=0 \text{ AND } \text{corpus}(x)=\text{no} \}|$$

Every FN refutes necessity. Every FP refutes sufficiency. This is the whole translation from
.the logical vocabulary into a measurable one

The figures below are read from the corpus at build time: **372 cases** (289 positive · 83 negative) across **34 papers**. The base rate is 78% positive — hence **MCC and not F_1** : a classifier that says yes
.to everything obtains a high F_1 and an MCC of zero

MCC	TN	FN	FP	TP	Definition
0.700+	70	29	13	260	condition-based definition (object/action, intention, values)
0.477+	72	87	11	202	the aesthetic definition
0.392+	81	150	2	139	art as expression
0.252+	33	44	50	245	broad
0.251+	82	216	1	73	narrow
0.238+	77	195	6	94	mimesis
—	83	2	0	287	circular
—	67	23	16	266	the cluster account
—	79	47	4	242	the historical definition
—	81	69	2	220	the institutional theory
—	59	7	24	282	open

Note the row for the circular control. It is nearly perfect, because it copies the answer. This
.is why circularity must be a **gate** and not a score: no statistical measure will catch it

What the matrix covers, and what it does not

Three of the six criteria are computed from these four cells: coverage ($\text{recall} = \text{TP}/(\text{TP}+\text{FN})$), discrimination ($\text{precision} = \text{TP}/(\text{TP}+\text{FP})$), and prediction — the same four cells on a held-out sample not used in constructing the definition. The remaining three lie outside it. **Typicality ranking** requires an order rather than labels, and is measured by rank correlation between the ranking a definition predicts and frequency of attestation in the corpus. **Applicability** is measured by κ . **Minimality** is a property of the definition rather than of the cases, and is measured by .ablation: remove one condition, re-score; a condition whose removal changes nothing is ornament

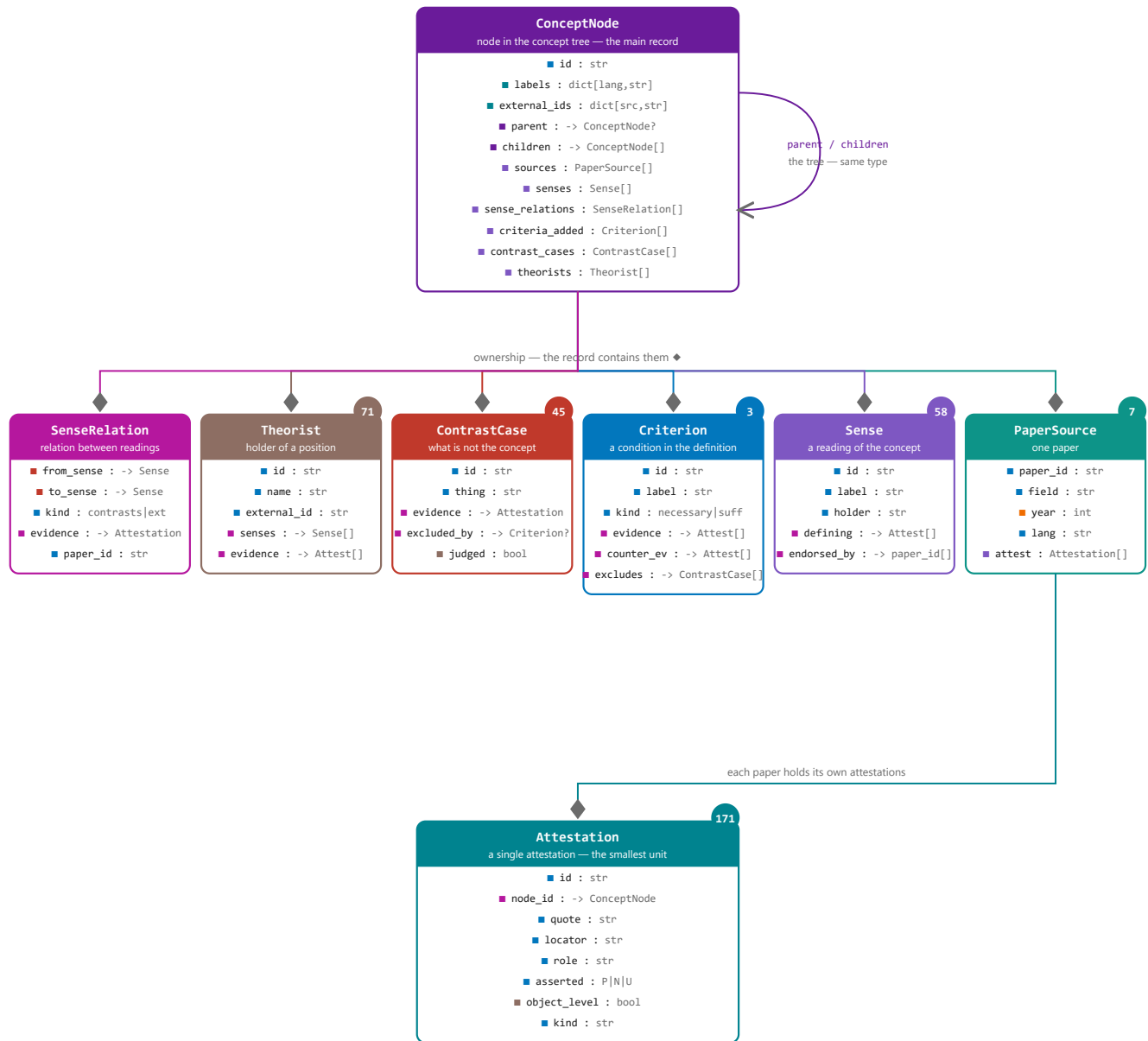
Three states, not two

A case is positive when a paper states explicitly that it is the concept, and negative when a paper states explicitly that it is not. The third state is **silence**, and **silence is not negation**. If silence were counted as negative, every paper that did not mention the Mona Lisa would be recorded as claiming that it is not art. This is the error that destroys measurements of this kind, and it is very .easy to make when counting automatically

The unit of analysis is accordingly neither the paper nor the passage but the **(case, paper) pair**. The same case may be positive in one paper and negative in another, and that disagreement is the most valuable datum in the corpus rather than noise: collapsing to the paper loses it, collapsing to the case loses who said it. Variance is decomposed with the paper as a random effect ($\text{ICC} = \text{between} / (\text{between} + \text{within})$), so that twenty-five attestations from a single paper .inflate nothing

Structure of the concept record .9

This is the record the measurement rests on. **Every field in the diagram is a link**; following it reaches that field's entry in the glossary in Appendix B, which states what the field does, what it .holds, and its data structure (string · number · array · dictionary · record · tree · graph)



.Rectangle = record · solid arrow = ownership · open arrow = reference. Every row links to the glossary

Reliability of application — κ .10

Applicability is not one more row in the table: **it is a precondition for everything else**. If two people applying the same definition to the same case reach different answers, the four cells of the .matrix measure the applier rather than the definition

We measured this directly. A blind sample of **140** (case × definition) pairs across seven definitions was coded by a second coder that was not the system which produced the first coding. The

n	Agreement	κ	Definition
20	90%	0.800+	the historical definition
20	75%	0.500+	condition-based definition (object/action, intention, values)
20	75%	0.500+	the institutional theory
20	75%	0.479+	the aesthetic definition
20	70%	0.417+	the cluster account
20	80%	0.412+	mimesis
20	85%	0.348+	art as expression
140	78.6%	0.547+	Overall

The hypothesis was refuted, and that is the finding. We stated in advance, in the coding sheet itself, that we expected κ for the cluster account to **collapse** relative to the others, because it does not fix how many of its ten criteria count as "enough". It did not collapse — +0.417, only +0.090 from the mean of the others. To the extent this sample allows, Gaut is .right: the threshold converges in practice even when it is left unstated

What is of concern. An overall κ of 0.547 is **moderate** rather than substantial on the conventional scale (Landis & Koch, 1977). Two coders disagree on roughly a quarter of the judgements on which every ranking in this document rests. For comparison, on a controlled layer of statistical terms we measured $\kappa = 0.767$ with the identical apparatus. **The difference is not noise: it is the cost of applying a verbal definition as against matching a term .from a closed list**, and it is the open problem of the field

One incidental result deserves recording. The definition with the highest agreement is the one disqualified at the recursion gate. "Intended for regard as prior works were regarded" is easy to agree on precisely because the test is simple to apply and demands no substantive judgement. **A .stable definition is not thereby a good one**, and the two measures must be reported side by side

The community: work in digital humanities, and how this connects .11

This work does not stand alone. A 2026 summer school at the Open University of Israel brought together eight researchers who all address one problem in different guises: **how to turn text into .data without losing what makes it text**. The account below is written from reading their papers

Reliability		Platform	Vocabulary	Unit of annotation	Researcher
	not measured	DistilBERT + Gephi	closed: 103 topics	a figurative instance — a manually extracted sub-sentential span	Gutman figurative language as a network
	not measured, by stated policy	Google Forms	mixed: questionnaire	the whole novel — a work-level record	Dekel public distant reading
	—	Hebrew corpus	corpus queries	an occurrence of an idiom in a corpus	Melnik idioms and their flexibility
	—	CATMA → ViS-À-ViS	hierarchical tagset	a span in the text	Marienberg-Milikowsky interpretive tagging
	—	TEI-Publisher	the TEI-XML standard	an entity in the text: person, place, subject	Sagiv Hasidic stories
	not measured	WHG · Recognito · Wikidata	closed: gazetteers	a pair of records , not a span in text	Rusinek MEHDIE · Kima

Reliability		Platform	Vocabulary	Unit of annotation	Researcher
	$\kappa = 0.82$	DictaLM 2.0	closed: two-stage scheme	the sentence	Rivlin-Angert political delegitimisation

What is shared, and what differs

Shared: all of them tag *a phenomenon in a text* — a metaphor, a place name, delegitimisation, an idiom. **Different:** for them tagging is **the object**; here it is **the substrate** — the evidence against which a definition is tested. The question asked is not "what is in this text" but "which definition of .the concept withstands this corpus

Two weaknesses recur across nearly all of this work, and they are precisely where the .present method contributes

Categories without definitions. Gutman's 103 topics carry no definition, and he concedes · 1 the consequence in a footnote: two interpreters may assign different topics to the same word, and he defers the reliability measurement to future work. 2 · **Reliability that is not measured.** Of the eight, only Rivlin-Angert measured it — three coders, a 170-sentence sample, $\kappa = 0.82$. For Dekel non-measurement is an explicit methodological position; in the MEHDIE paper there is no accuracy figure of any kind, and the evidence offered is a single .worked case

This is what the present method adds: a mandatory verbatim quotation for every tag, and a .measurable definition for every category

Why it works there and breaks here

Rusinek's Kima has the same shape as the record used here — place → name variant → dated attestation against concept → wording → quotation . The architectures are identical. The only difference is that a place has a **referent**: one can transliterate it to IPA and check whether two spellings sound the same. An abstract concept admits no such test. .That is the gap, and it is the open problem for all of us

Interoperability — what must be emitted

:Connection requires no agreement between projects. It requires three format decisions

An opaque identifier instead of a label. Where the tag is the word, any change of wording •
silently breaks every external link. A stable identifier, with the words hanging off it as

. `skos:altLabel` `skos:prefLabel` and

A textual anchor that is not an offset. A `TextQuoteSelector` (quotation plus preceding and •
following context) survives re-editions; character offsets do not. Offsets for tools that need them
.are derived from it at export time

Resolution to an external authority — Wikidata, GeoNames. Once two layers point at the same •
.identifier they join without any bilateral arrangement

Two export formats carry the community's actual practice: two-tier **TEI** (metadata in `teiHeader` ,
.span annotations in `standOff`), and a pair of **Gephi** CSV files (nodes and edges)

And in the other direction: defining Gutman's 103 categories with this instrument would
supply the reliability measure his work lacks, and would supply this work a test case that is not
its own — which is the more valuable half of the exchange, because on that corpus the
.authors hold no competing definition

Limitations .12

This list belongs in the body of the text and not in a footnote. A new measure that arrives
with its limitations is received better than one that arrives without them, and a reviewer will
.find them regardless

Reliability is moderate, and it bounds everything else. $\kappa = 0.547$ over 140 blind pairs. Two •
.coders disagree on roughly a quarter of the judgements the rankings rest on

Comparison is valid within a single run only. The same definition scored +0.714 in one run •
and +0.591 in another on identical cases; the only difference was the wording of the instruction
given to coders. MCC is not a property of a definition but of the pair (definition × instruction).
.Numbers from two tables must therefore never be compared

- **The negatives are not a representative sample.** They are contrast cases someone troubled to write down, not a random sample of what is not the concept. Accuracy and specificity are therefore inadmissible, and the corpus was deliberately extended with hard negatives — .advertisements, games, sport — where permissiveness is caught
- **.There is no access to authorial intent.** What was not written is not measured
- **Recall has a ceiling.** What was missed cannot be known without knowing the whole set; .coverage is therefore coverage relative to a documented pool, and must be named as such
- **The corpus is still small.** 34 papers. The bootstrap unit is the paper, so the effective n is 34 .rather than 372 — which is why the lower places in a ranking are not separable
- **.The Wittgensteinian position cannot be scored,** as stated in §5

Appendix A — formulae

All measures are computed from the four cells. TP and FN are counted over cases the corpus adjudicated positive, FP and TN over those it adjudicated negative

$$\begin{aligned} \text{discrimination — precision} &= TP / (TP + FP) \\ \text{coverage — recall} &= TP / (TP + FN) \\ F1 &= 2 \cdot \text{precision} \cdot \text{recall} / (\text{precision} + \text{recall}) \\ MCC &= (TP \cdot TN - FP \cdot FN) / \sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)} \end{aligned}$$

Why MCC and not F_1 . F_1 ignores TN, so on a positively skewed corpus a classifier that says yes to everything obtains a high F_1 . MCC gives that same classifier exactly **0.000** (Matthews, 1975; Chicco & Jurman, 2020). F_1 is reported beside it, never alone

Reliability

$$\kappa = (p_o - p_e) / (1 - p_e)$$

observed agreement · p_e = agreement expected by chance from the two coders' = p_o marginals (Cohen, 1960). Interpretation of values follows Landis & Koch (1977). For more than two coders or mixed scales, Krippendorff's alpha (Hayes & Krippendorff, 2007)

Variance between papers and within them

:The same case may be judged in several papers. Variance is decomposed by analysis of variance

(Shrout & Fleiss, 1979) $ICC = \sigma^2_{\text{between}} / (\sigma^2_{\text{between}} + \sigma^2_{\text{within}})$

High ICC — papers disagree with one another and are internally consistent → the .disagreement is cross-disciplinary, and the senses should be split

Low ICC — the same paper holds several positions → the concept is contested within a single .author, which is a finding in itself

This also answers "count cases or count papers": a hierarchical model with the paper as a random .effect counts both, and twenty-five attestations from one paper inflate nothing

Confidence intervals and comparison

Confidence intervals are computed by **bootstrap** (Efron, 1979) **at the level of the paper and not the case**, because cases from one paper share a single author's usage and are not independent; resampling cases yields intervals that are too narrow and a significance that does not exist. Two definitions on the same cases are compared with a paired test (McNemar, 1947), and a tie is .declared when neither beats the other in 95% of resamples

Appendix B — field glossary

the concept record **ConceptNode**

The record that holds one concept. Everything we know about "art" hangs from it. It is also a node .in a tree, so it can be split into sub-types without duplicating information

str **STRING** **id**

.What it does: The concept's only key in the system

What it holds: A short, fixed string, e.g. art (text)

dict[lang,str] **DICTIONARY** **labels**

.What it does: The concept's name in each language

What it holds: A dictionary: language code → name. he→omanut, en→art (key→value pairs)

`dict[src, str]`**DICTIONARY****external_ids**

What it does: Links to authority records outside the project, so the layer can join to Wikidata and .to MEHDIE

What it holds: A dictionary: authority name → its identifier. wikidata→Q735 (key→value pairs)

`-> ConceptNode?`**TREE****parent**

What it does: The broader concept this one derives from. For "painting" the parent is "art

What it holds: A pointer to a record of the same type, or empty if this is the root (reference to a record of the same type: parent or children)

`-> ConceptNode[]`**TREE****children**

What it does: The sub-types. This is what allows a concept to be split into ten sub-types quickly, with the children inheriting the parent's criteria by reference rather than by copy

What it holds: A list of pointers to records of the same type (a list of items of one type)

`PaperSource[]`**ARRAY****sources**

What it does: The papers that mention the concept, grouped by paper -- because twenty-five mentions in one paper are one opinion, not twenty-five

What it holds: A list of PaperSource records (a list of items of one type)

`Sense[]`**ARRAY****senses**

What it does: The different readings the concept receives. One record can hold contradictory senses without averaging between them

What it holds: A list of Sense records (a list of items of one type)

`SenseRelation[]`**GRAPH****sense_relations**

What it does: Who contradicts whom, who extends whom -- the concept's internal graph

What it holds: A list of SenseRelation records (a list of items of one type)

`Criterion[]`**ARRAY****criteria_added**

What it does: The conditions this concept adds beyond its parent's. Inheritance is resolved by walking upward, so a change at the parent reaches every child immediately

What it holds: A list of Criterion records (a list of items of one type)

`ContrastCase[]`**ARRAY****contrast_cases**

What it does: What is not the concept. Without these, discrimination cannot be measured -- only .coverage

What it holds: A list of ContrastCase records (a list of items of one type)

Theorist[] **ARRAY** theorists

What it does: Who holds which reading

What it holds: A list of Theorist records (a list of items of one type)

one paper **PaperSource**

One paper that mentions the concept, with all of its attestations inside it. This is what stops twenty-five mentions in a single paper from counting as twenty-five independent opinions: they are one .author's position

str **STRING** paper_id

What it does: Which paper this is

What it holds: A stable identifier for the paper (text)

str **STRING** field

What it does: Its discipline -- so that crossing of fields can be checked

What it holds: A discipline label (text)

int **NUMBER** year

What it does: When it was published -- so that change over time is visible

What it holds: A year (number)

str **STRING** lang

What it does: The language of the source

What it holds: A language code (text)

Attestation[] **ARRAY** attest

What it does: Every attestation from this paper

What it holds: A list of Attestation records (a list of items of one type)

a reading of the concept **Sense**

One reading the concept receives. A single record can hold contradictory senses without averaging
.between them, which is the point: disagreement is data

str **STRING** **id**

.What it does: The sense's key

What it holds: A short string (text)

str **STRING** **label**

.What it does: What this sense is called

What it holds: A short phrase (text)

str **STRING** **holder**

.What it does: Who holds it

What it holds: A theorist's name or identifier (text)

-> Attest[] **ARRAY** **defining**

.What it does: The quotations the sense is derived from -- not our paraphrase of them

What it holds: A list of pointers to attestations (reference to another record, not a copy of it)

-> paper_id[] **ARRAY** **endorsed_by**

.What it does: Which papers hold this sense

What it holds: A list of paper identifiers (a list of items of one type)

a condition in the definition **Criterion**

.One condition a definition asserts, carrying both the evidence for it and the evidence against it

str **STRING** **id**

.What it does: The condition's key

What it holds: A short string (text)

str **STRING** **label**

.What it does: The condition in words

What it holds: A phrase (text)

necessary|sufficient **CLOSED LIST** **kind**

What it does: Whether it is asserted as necessary or as sufficient -- two things that are measured differently

What it holds: One of a closed list of values

-> Attest[] **ARRAY** **evidence**

What it does: Quotations that support the condition

What it holds: A list of pointers to attestations (reference to another record, not a copy of it)

-> Attest[] **ARRAY** **counter_ev**

What it does: Quotations that refute it. Every item here is a refutation of necessity

What it holds: A list of pointers to attestations (reference to another record, not a copy of it)

-> ContrastCase[] **ARRAY** **excludes**

What it does: Which contrast cases this condition filters out

What it holds: A list of pointers to contrast cases (reference to another record, not a copy of it)

what is not the concept **ContrastCase**

A case the literature states is NOT the concept. Without these, discrimination cannot be measured at all -- only coverage

str **STRING** **id**

What it does: The case's key

What it holds: A short string (text)

str **STRING** **thing**

What it does: What the case is, briefly

What it holds: A short description (text)

-> Attestation **RECORD** **evidence**

What it does: The quotation it was taken from

What it holds: A pointer to one attestation (reference to another record, not a copy of it)

-> Criterion? **RECORD** **excluded_by**

What it does: Which condition filtered it out -- this is how one learns which condition is doing .work

What it holds: A pointer to one criterion, or empty (reference to another record, not a copy of it)

bool **BOOLEAN** **judged**

What it does: Whether the paper adjudicated it at all. An unadjudicated case is not counted as a .negative; it goes into the unresolved rate

What it holds: Yes or no

holder of a position **Theorist**

.Who holds which reading. A node in the graph, not metadata

str **STRING** **id**

What it does: Internal key

What it holds: A short string (text)

str **STRING** **name**

What it does: The name as it appears

What it holds: A name (text)

str **STRING** **external_id**

What it does: Wikidata -- so that "Arthur Danto" and "Arthur C. Danto" are not counted twice

What it holds: An authority identifier (text)

-> Sense[] **ARRAY** **senses**

What it does: Which readings this person holds

What it holds: A list of pointers to senses (reference to another record, not a copy of it)

-> Attest[] **ARRAY** **evidence**

What it does: Where we know it from

What it holds: A list of pointers to attestations (reference to another record, not a copy of it)

relation between readings **SenseRelation**

.A weighted edge between two senses: who contradicts whom, who extends whom

-> Sense **RECORD** **from_sense**

.What it does: One end of the edge

What it holds: A pointer to a sense (reference to another record, not a copy of it)

-> Sense **RECORD** **to_sense**

.What it does: The other end

What it holds: A pointer to a sense (reference to another record, not a copy of it)

contrasts | extends | narrows **CLOSED LIST** **kind**

.What it does: Contradicts, extends, narrows -- and in future a weight on each edge

What it holds: One of a closed list of values

-> Attestation **RECORD** **evidence**

.What it does: The quotation that grounds the relation

What it holds: A pointer to one attestation (reference to another record, not a copy of it)

str **STRING** **paper_id**

.What it does: Who asserts this relation

What it holds: A paper identifier (text)

a single attestation **Attestation**

One mention in one paper, with the verbatim sentence it came from. This is the atom of the whole
.structure

str **STRING** **id**

.What it does: The attestation's key

What it holds: A short string (text)

-> ConceptNode **RECORD** **node_id**

What it does: Which concept or sub-type it belongs to. Always the most specific node that fits,
.otherwise information smears upward

What it holds: A pointer to a concept record (reference to another record, not a copy of it)

str **STRING** quote

.What it does: The verbatim sentence from the paper. Must be an exact substring

What it holds: The sentence itself (text)

str **STRING** locator

.What it does: Where in the text -- page or offset

What it holds: A locator (text)

str **CLOSED LIST** role

.What it does: What the mention does: defines, exemplifies, refutes, merely mentions

What it holds: One of a closed list of values

P|N|U **CLOSED LIST** asserted

What it does: Yes / no / not adjudicated. The third value is what allows us to be honest about
.what the literature did not decide

What it holds: One of three values

bool **BOOLEAN** object_level

What it does: Whether this is an instance of the concept or a claim about it. Conflating the two
.destroyed an entire run

What it holds: Yes or no

str **CLOSED LIST** kind

What it does: Artefact, performance, text, natural item, event, abstract. This is what makes it
.possible to define sub-concepts such as "an object" from the corpus rather than by assumption

What it holds: One of a closed list of values

Appendix C — when the method proposes a definition itself

The apparatus does not only rank definitions submitted to it; it can assemble one from atomic conditions drawn from the definitions already in the corpus, and score it like any other. This is what
.a user asks for when they have no candidate of their own

Here the important lesson about order of operations was learned. In the first run the search was allowed to see the full inventory of conditions, including one phrased as "produced with the intention of making a work of art". The search selected exactly that one, obtained the highest score in the table, and had to be struck out afterwards — a fully circular definition that the machine rediscovered on its own within seconds. Circular conditions are now flagged **in the data itself** and removed from the vocabulary **before** the search begins. **A .gate applied after generation is a gate applied too late**

A second lesson, general and not specific to any one concept. In a later run the method proposed a definition resting on the property "exhibits an individual point of view". It scored highly — until seven negative control cases were added (an opinion column, a party manifesto, a closing argument in court, a scientific paper, a book review, a restaurant review, a .personal blog post). The definition admitted **all seven**, and the score collapsed

The method was not at fault. The corpus did not contain what could have taught it. No academic paper writes "an opinion column is not a work of art", because no one needs to. Hence the principle, which holds for any concept: **a definition cannot be falsified by a .corpus that does not contain what would falsify it**

The control cases are held in a separate file and marked explicitly as the researcher's labelling rather than the literature's, and every score is reported with and without them. They are not evidence; they are an instrument of control, exactly like the deliberately broken definitions run .alongside the real ones

Appendix D — reproduction

:Every number in this document is produced from the data files and can be recomputed

```
python TOOLS/final_ranking.py ART_CASE/cases_v3.json ART_CASE/art_v7_verdicts.json •
python TOOLS/cluster_k_sweep.py then python TOOLS/cluster_criteria.py •
python TOOLS/make_defs_kappa.py score KAPPA_STUDY_DEFS/responses.csv •
this document — python TOOLS/build_defs_preprint_en.py •
```

0	9	8	7	6	5	4	3	2	1	k
0.576+	0.660+	0.738+	0.781+	0.847+	0.882+	0.826+	0.770+	0.620+	0.282+	MCC

.Best fixed threshold $k = 5$ · worst $k = 1$ · range 0.6 MCC · with k chosen per case, MCC 1.0

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